

Counterfactual Explanations for Anomalies in Photovoltaic Power Forecasting

Bodo Wirth

bodow@stanford.edu



Introduction

- **Solar energy is vital for renewable energy adoption**, but fluctuations in photovoltaic power output hinder reliable grid integration.
- Power output prediction via deep learning is necessary, but it is difficult to interpret and trust results, especially in the case of anomalies.

This project uses a state of the art neural network to forecast PV power output from a variety of physical and meteorological factors before applying **counterfactual explanations** to interpret two types of negative anomalies:

- **Type 1 Anomalies:** Model overpredicts power output significantly.
- **Type 2 Anomalies:** True power output drops below the expected range and the model predicts this drop.

Data

- 2020 and 2021 power output and physical data collected hourly from NREL 75-kw solar field in Golden, Colorado.
- Aligned weather forecast data and added Gaussian noise to improve the generalizability of the model.
- Data normalised for all models and PCA performed before generating the counterfactual explanations.

Features

Raw Inputs: Weather data, solar panel conditions, and power output. See Figure 1.

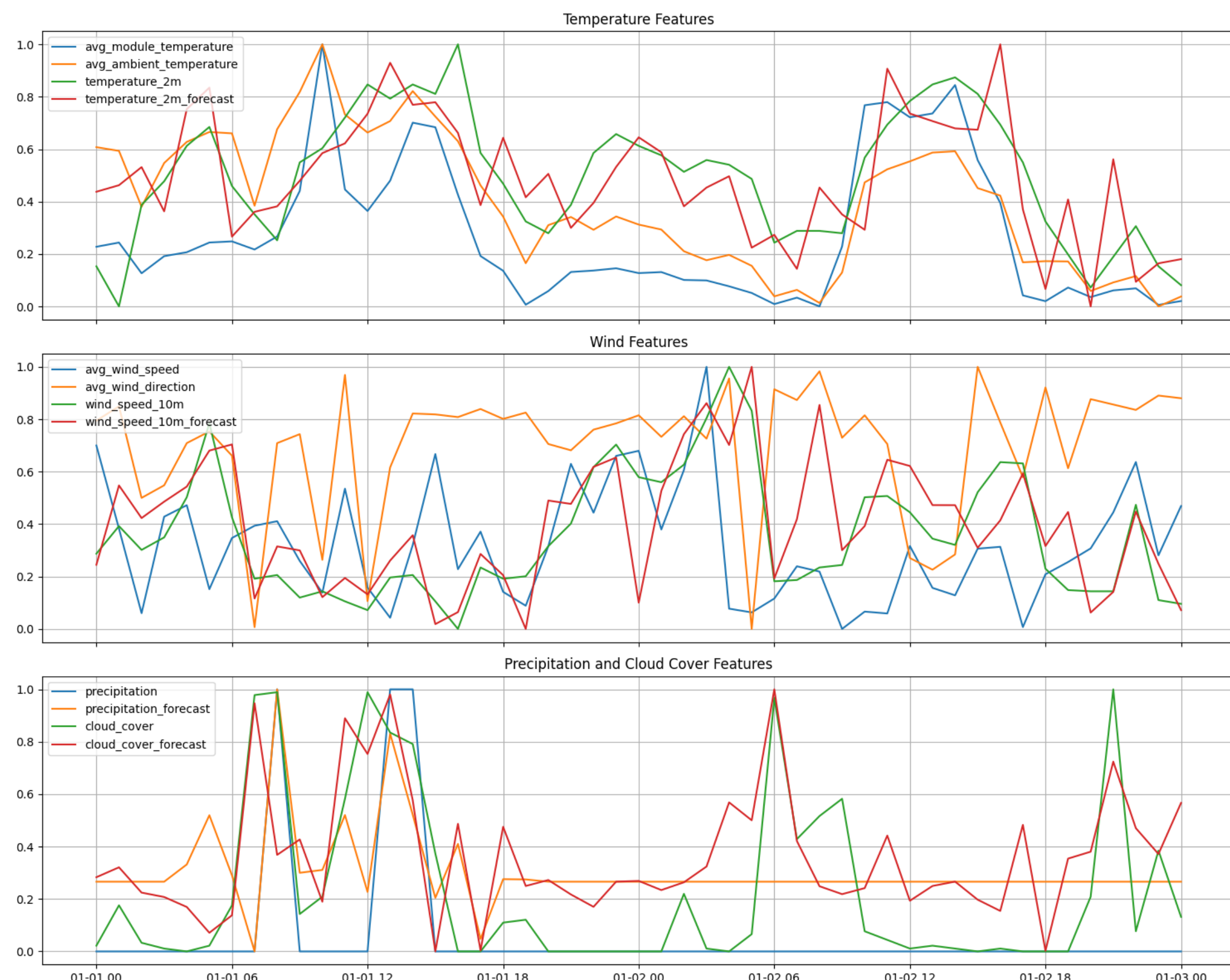


Figure 1. Normalized Features

Derived Features: Time-based features transformed into sine and cosine to capture cyclical patterns and PCA for dimensionality reduction.

Models

Linear Regression: Baseline model for comparison.

FDNet: Advanced neural network optimized for time series forecasting utilising:

- *2D Convolutions:* Extract temporal and variate features.
- *Focal Input Sequence Decomposition:* Handles long-sequence inputs by dividing them into sub-sequences with differing granularity with increased attention to those in closer temporal proximity.
- *Decomposed Forecasting:* Improves robustness to non-stationary time series such as the seasonal changes that might occur in my data.

Counterfactual Explanations (CEs): Explore "what if" scenarios by finding minimal changes to the input $\mathbf{x}$ such that the model's output $f(\mathbf{x})$ changes to a desired target $\mathbf{y}^*$. By analyzing which features were most altered in order to change a classification, one can learn which are most significant to the model's predictions, while the direction of the change can indicate in what sense they affect these forecasts.

Results

Prediction Performance Metrics:

Model	Time Span (h)	MSE	MAE	R^2
Linear Regression	1	1.408	0.758	0.758
	12	1.972	1.009	0.662
	48	1.956	0.941	0.665
FDNet	1	0.019	0.181	0.982
	12	0.202	0.562	0.809
	48	0.262	0.670	0.753

Table 1. Performance metrics for Linear Regression and FDNet.

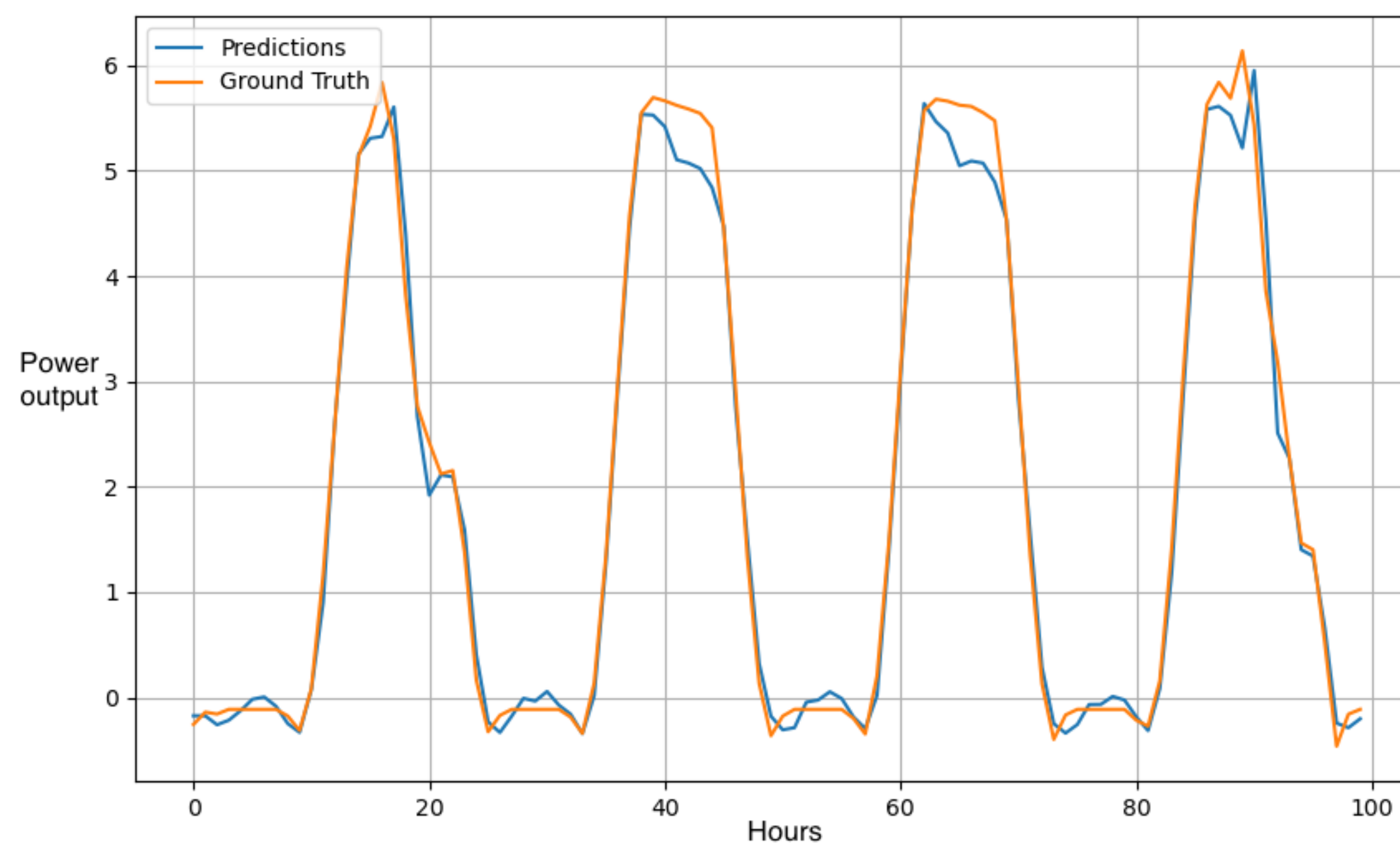


Figure 2. Predicted (1 Hour) and True Power Output Sample

As expected, FDNet significantly outperformed Linear Regression across all time spans, particularly for short-term predictions. For longer time spans, FDNet's performance declined, suggesting potential overfitting or inherent issues in the model architecture, or my implementation.

Results (Continued)

Counterfactual candidates provide feature changes required to shift data points from anomalous to not.

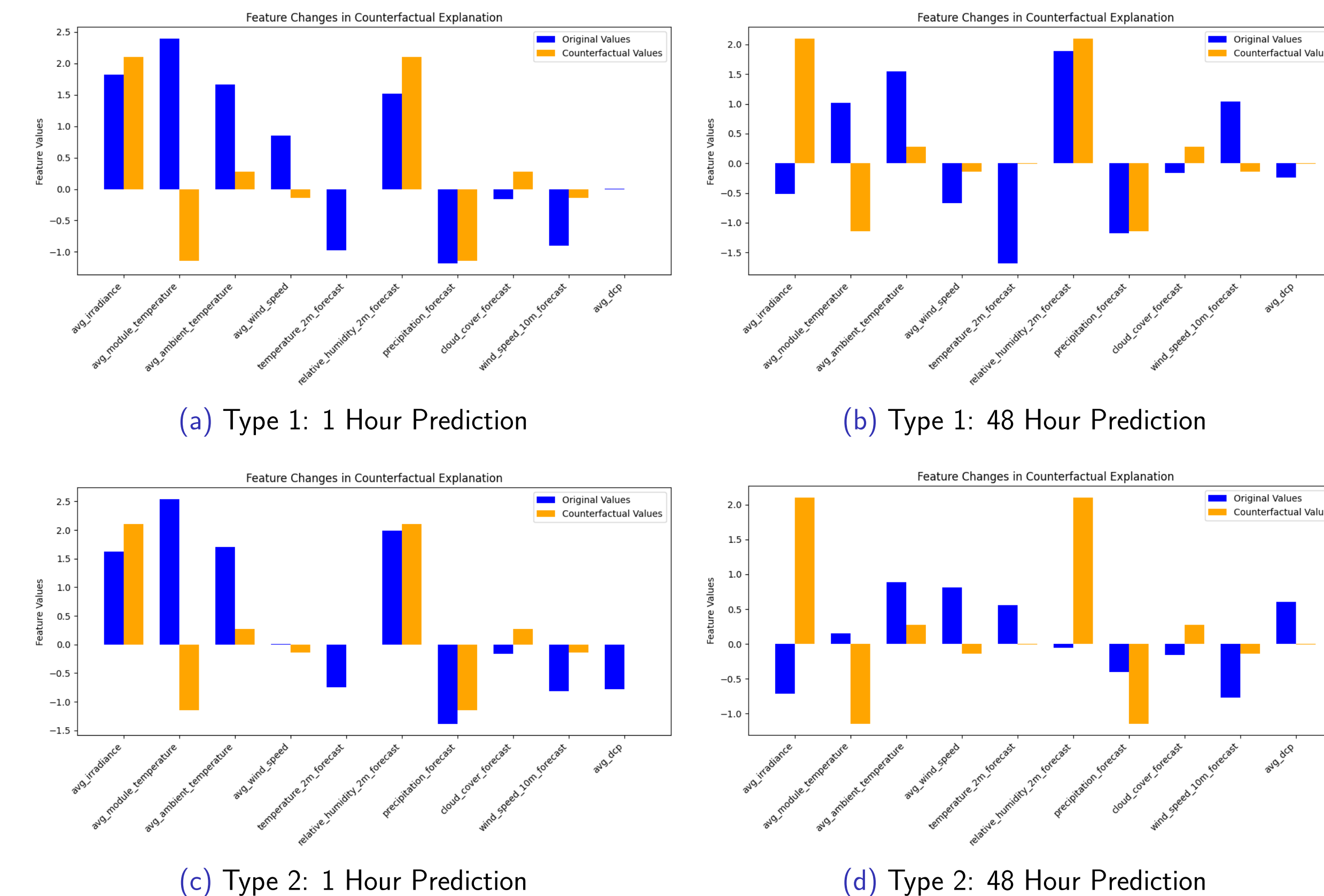


Figure 3. Feature changes in counterfactual explanations for Type 1 and Type 2 anomalies.

Type 1 Anomalies: Temperature-based features most significant. For 48-hour predictions, irradiance also played a dominant role.

Type 2 Anomalies: For 1-hour predictions, panel temperature was overwhelmingly the key factor, agreeing with our knowledge of module efficiency being correlated with lower temperatures. For 48-hour predictions, irradiance and humidity emerged as important features. This disagrees with the consensus of the impact of humidity on PV output.

Discussion

- FDNet provided excellent performance for short-term forecasts but struggled with long-term dependencies, highlighting potential data or model limitations that the CEs corroborated with the model's use of unexpected relationships (e.g. humidity effects).
- Counterfactual explanations revealed critical features influencing anomalies. Provide actionable insights into improving model accuracy and potentially PV output (regulate module temperature).

These results demonstrate the utility of counterfactual explanations in helping to improve model trust, interpretability, and operational decision making in solar farm management.

Future Work

- Combine counterfactual explanations with SHAP to gain deeper insights into feature contributions and interactions.
- Test counterfactual-based recommendations in simulated solar farms to assess their impact on power output optimization and potential real-world use cases.